

3 R intro; Epidemic Data and Time Series Tools

4 Epidemic Data Tools II; Mechanistic Modelling of Recurrent Epidemics I



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 4MB3/6MB3 Mathematical Biology

Instructor: David Earn

Lecture 3

R intro; Epidemic Data and Time Series Tools
Tuesday 22 September 2026

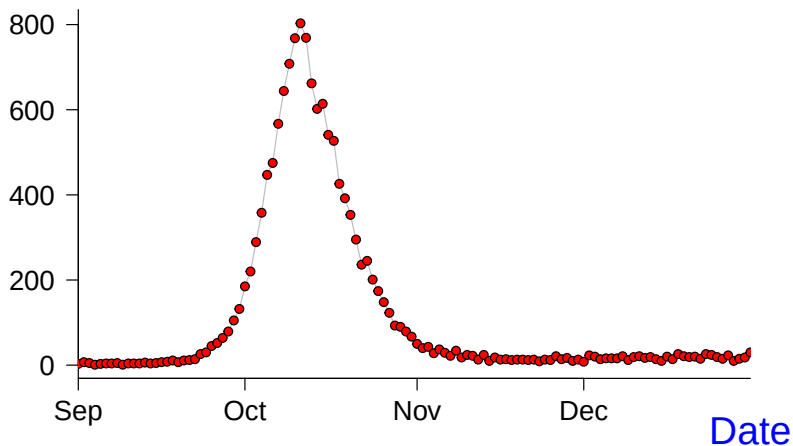
Announcements

■ **Assignments:**

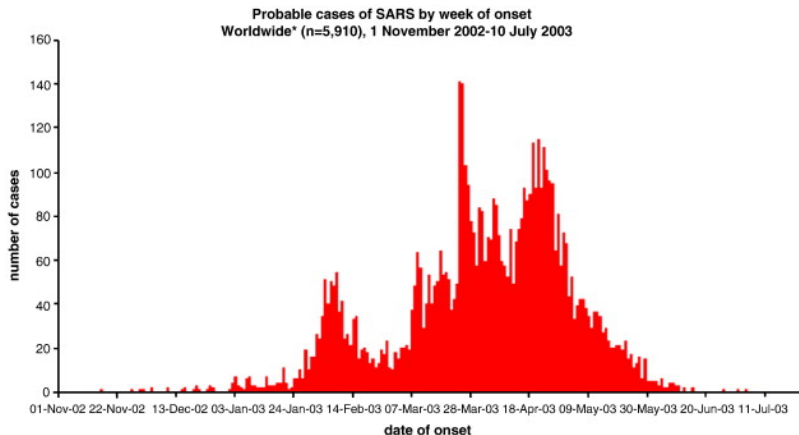
- Assignment 1 due Thursday 24 September 2026 at 11:59 p.m.
- Assignment 2 due Thursday 8 October 2026 at 11:59 p.m.
(good to work on before class on Tuesday 29 September 2026)

P&I Mortality, Philadelphia, 1918

P&I Deaths

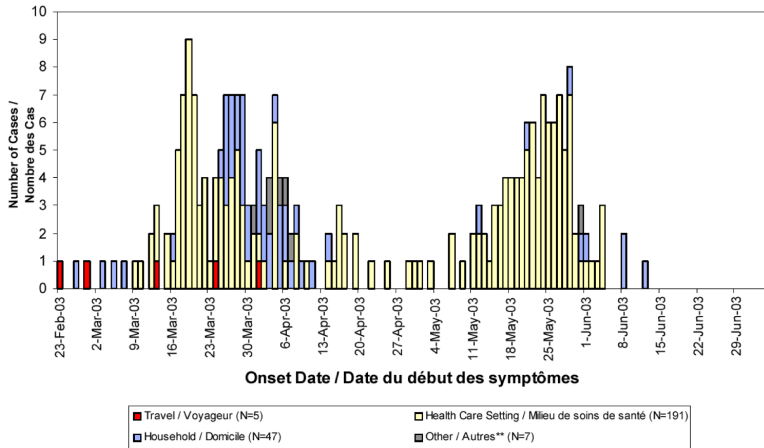


SARS-CoV-1 in 2003 (Worldwide)



*This graph does not include 2,527 probable cases of SARS (2,521 from Beijing, China), for whom no dates of onset are currently available.

SARS-CoV-1 in 2003 (Toronto)



$N = 249$ (of 250 reported)

Some SARS-CoV-1 Facts

- High case fatality
 - 1918 flu $< 3\%$
 - SARS-CoV-1 $> 10\%$
- Long hospital stays
 - Mean time from admission to discharge or death:
~ 25 days in Hong Kong
- 8098 probable cases, 774 deaths
- How bad would it have been if it had not been controlled?

COVID (ancestral) hospitalization and survival in Ontario

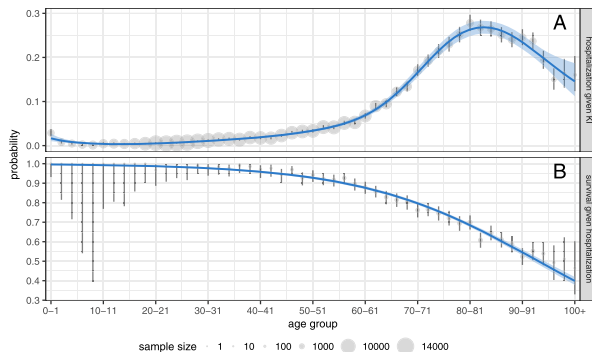
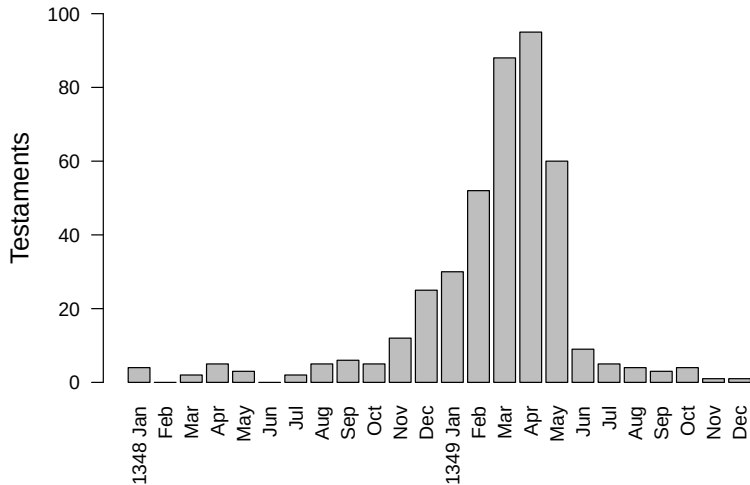


Fig. 4 Age-dependent COVID-19 hospitalization probability for known SARS-CoV-2 infection (panel **a**) and survival probability for hospitalized patients (panel **b**) in Ontario. We give age-by-age estimates of each probability (points; 95% exact binomial confidence intervals given by vertical lines), where point area is proportional to age-specific sample size. We additionally provide more precise estimates of these probabilities under stricter assumptions, modelling the hospitalization probability using a generalized additive model and the survival probability using a generalized linear model (curves; 95% confidence bands given by shaded regions). See "Methods" section for details

Papst et al (2021), BMC Public Health, 21:706

The Black Death in London, England, 1348–1349



London Bill of Mortality, 26 Sept to 3 Oct 1665

The Diseases and Casualties this Week.		London 41. From the 26 of September to the 3 of October. 1665	
	Frighted	1	1
	Goat	1	1
A Borne Aged Ague Apoplexie Childred Chromes Cold Consumption Convulsion Cough Drople Drowned at St. Martin in the Fields Feaver Fistula Flox and Small-pox Flux Found dead in the Fields at St. Mary Islington	Grief	1	1
	Gripping in the Guts	35	35
Jaundies Impothame Infants Kingsevil Meagrome Plague Purples Rickets Rising of the Lights Rupure Scurvy Spotted Feaver Stilborn Stone Stopping of the Stomach Suddenly Surfeit Teeth Thrush Tifick Vomiting Winde Wormes	Jaundies	35	35
	Impothame	2	2
	Infants	2	2
	Kingsevil	2	2
	Meagrome	2	2
	Plague	553	553
	Purples	2	2
	Rickets	2	2
	Rising of the Lights	2	2
	Rupure	13	13
	Scurvy	1	1
	Spotted Feaver	65	65
	Stilborn	10	10
	Stone	1	1
	Stopping of the Stomach	1	1
	Suddenly	6	6
	Surfeit	36	36
	Teeth	11	11
	Thrush	3	3
	Tifick	5	5
	Vomiting	4	4
	Winde	1	1
	Wormes	12	12
Males — 68 Females — 78 In all — 146		Males — 3212 Females — 3248 In all — 6460	
Buried — 553 Decreased in the Burials this Week — 1837 Parishes clear of the Plague — 7 Parishes Infected — 123		Plague — 553 Decreased in the Burials this Week — 1837 Parishes clear of the Plague — 7 Parishes Infected — 123	
The Affize of Bread (as forth by Order of the Lord Mayor and Court of Aldermans, A penny Wheaton Loaf to contain Nine Ounces and a half, and three half-penny White Loaves the like weight.		The Affize of Bread (as forth by Order of the Lord Mayor and Court of Aldermans, A penny Wheaton Loaf to contain Nine Ounces and a half, and three half-penny White Loaves the like weight.	

London Bill of Mortality, 26 Sept to 3 Oct 1665

Frighted	1
Gowt	1
Grief	3
Griping in the Gurs	35
Jaundies	2
Imposthume	8
Infants	9
Kingsevil	2
Meagrome	2
Plague	5533
Purples	2
Bickers	

Mortality Bills are typically handwritten

LONDON 29. From the 4. of July to the 11. of August 1665.											
Buried, Plag.			Buried, Plag.			Buried, Plag.			Buried, Plag.		
St Alban Woodstreet	2	1	St Clement Eastcheap	1		St Margaret Newfish			St Michael Crookedla	4	3
Alhallows Bark-			St Dionis Backchurch	2		St Margaret Patrons			St Michael Queenhit	2	
Alhallows Breadstreet	1		St Dunstons East			St Mary Abchurch	1		St Michael Quern		
Alhallows Great			St Edmund Lombardst			St Mary Aldermanbury			St Michael Royal		
Alhallows Honilane			St Ethelborough	2		St Mary Aldemary			St Michael Woodstreet		
Alhallows Lefi	1		St Faiths	1		St Mary le Bow			St Mildred Breadstreet		
Alhallows Lombardst			St Gabriel Fenchurch			St Mary Bothaw			St Mildred Poultry		
Alhallows Staining			St George Botolphane			St Mary Colechurch			St Nicholas Acon		
Alhallows the Wall	4	3	St Gregories by St Paul			St Mary Hill			St Nicholas Coleabby		
St Alphage			St Hellen	2	1	St Mary Mag. Milkstr			St Nicholas Olaves		
St Andrew Hubbard			St James Dukes place	1		St Mary Mag. Oldfish			St Olave Harcourt		
St Andrew Underhafe	3		St James Garlickhithe	1		St Mary Mounthaw			St Olave Jewry		
St Andrew Wardrobe	1		St John Baptst			St Mary Summerfet	2	1	St Olave Silverstreet	4	1
St Anne Aldergate	1		St John Evangelist			St Mary Stainings			St Pancras Soperlane		
St Anne Blackfryers	7	6	St John Zichary			St Mary Woolchurch			St Peter Cheap		
St Antholins Parish	7		St Katharine Coleman	1		St Mary Woolnoth			St Peter Cornhil		
St Austins Parish			St Katharine Creechur			St Martins Iremongerl			St Peter Paulwharf		
St Barthol. Exchange	1		St Lawrence Jewry			St Martins Ludgate	2	1	St Peter Poor	1	
St Bennet Fynck			St Lawrence Pountney			St Martins Orgars			St Steven Colemanstr	2	1
St Bennet Gracechurch	2		St Leonard Eastcheap			St Martins Outwich	1		St Steven Walbrook		
St Bennet Paulwharf	7		St Leonard Fosterlane			St Martins Vintrey			St Swithin	2	1
St Bennet Sherehog			St Magnus Parish	1		St Matthew Frydaystr			St Thomas Apostle	1	1
St Boroloph Billingsgate			St Margaret Lothbury			St Michael Bassishaw	5	4	Trinity Parish	1	
Christ Church	5	3	St Margaret Moses			St Michael Cornhil			St Vedast alias Fosters		
St Christophers											
Christened in the 9 Parishes within the walls			Buried			86			Plague		
St Andrew Holborn	00	40	St Boroloph Aldergate	11	9	St George Southwark	13	4	St Sepulchres Parish	117	81
St Bartholomew Great	4	4	St Boroloph Aldgate	24	4	St Giles Cripplegate	105	42	St Thomas Southwark	7	5
St Bartholomew Lefi			St Boroloph Bishopsgate	37	29	St Olave Southwark	20	6	Trinity Minories		
St Bridget	24	14	St Dunstan West	19	9	St Saviour Southwark	21	1	At the Pesthouse	6	6
Bridewel Precinct	1	1									
Christened in the 16 Parishes without the walls			Buried			473			Plague		
Christs Church			St Kath. near the Tower	5	1	St Mary Islington	3	2	St Paul Shadwel		
St John at Hackney	1		Lambeth Parish	21	13	St Mary Newington	16	5	Rotherhithe Parish	7	3
St Giles in the Fields	208	215	St Leonar d Shoreditch			St Mary Whitechappel			Stepney Parish	47	1
St James Clerkenwel	5		St Magdalen Bermond.	14							
Christened in the 12 Out-parishes in Middlesex and Surrey			Buried			455			Plague		

But handwriting is usually very clear

A historical document from the Great Plague of London, 1665, showing a table of burials and deaths by parish. The document is titled "LONNDRN 29th" and lists the following data:

	Buried.	Plag.
St Alban Woodstreet	2	1
Alhallowes Bark.	2	
Alhallowes Breadstreet	1	
Alhallowes Great —		

But handwriting is usually very clear

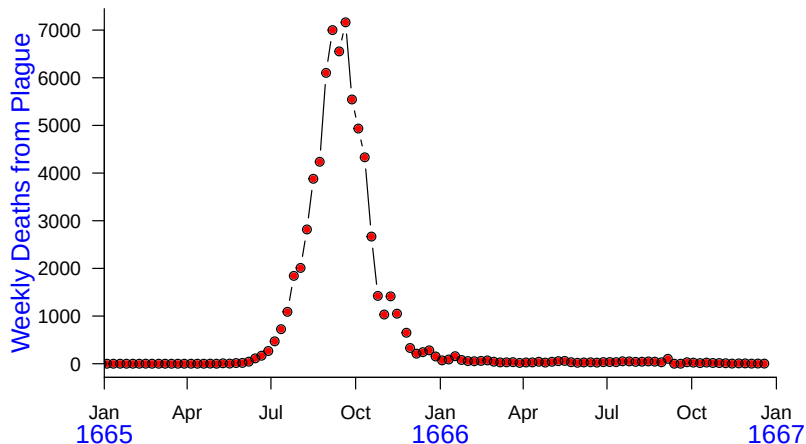
St Christopher's

Baptised in 97 the Parishes

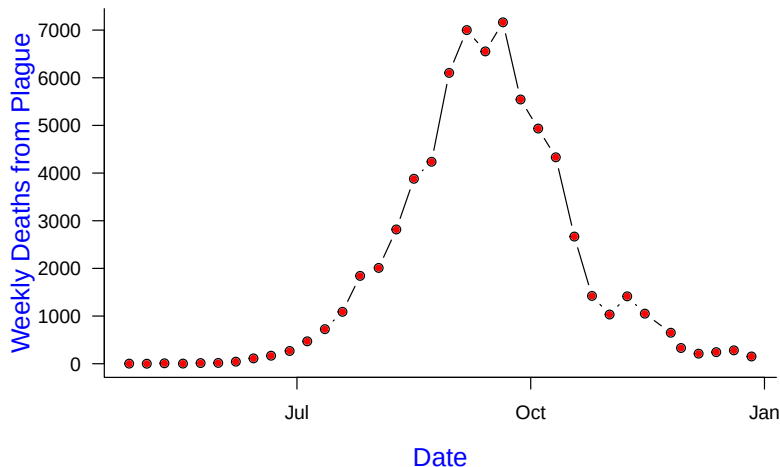
St Andrew Holborn—	66	40	St
St Bartholomew Great	7	4	St
St Bartholomew Less—			St
St Bridget—	24	17	St
Bridewel Precinct—	1	1	

Baptised in the 16 Parishes

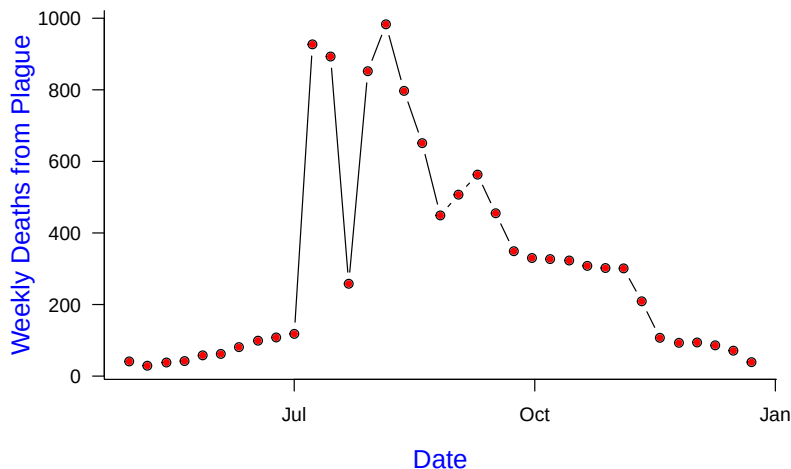
The Great Plague of London, 1665



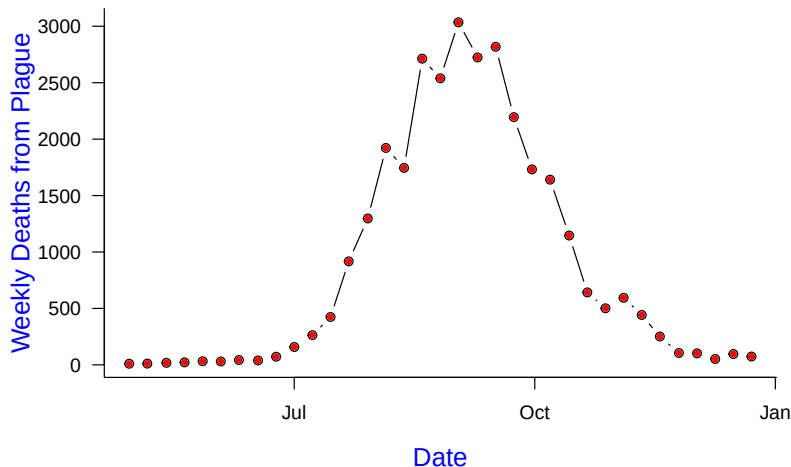
The Great Plague of London, 1665



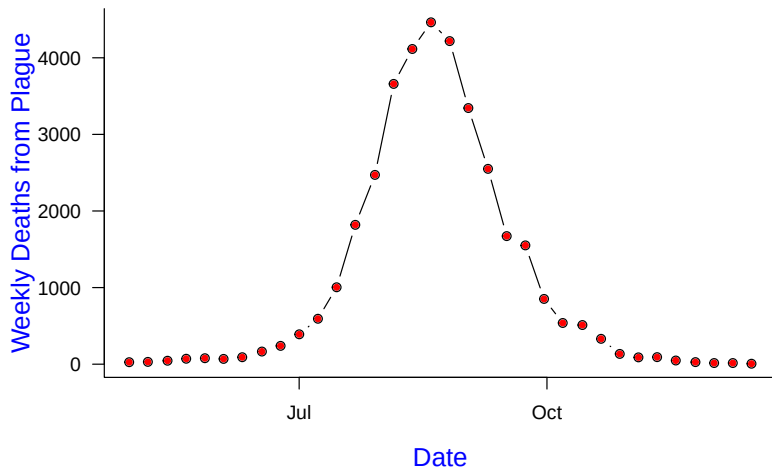
London Plague of 1593



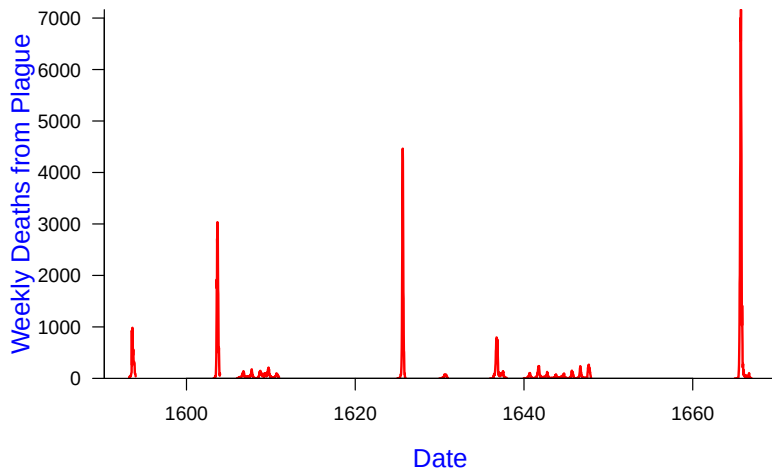
London Plague of 1603



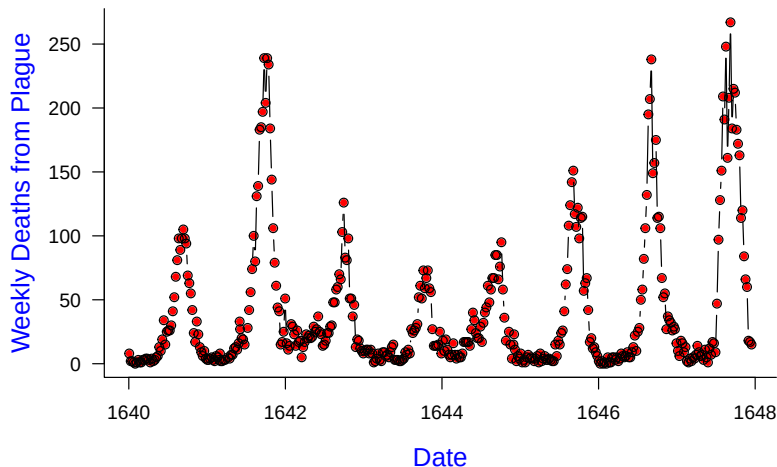
London Plague of 1625



Weekly Deaths from Plague in London, 1592–1666



Weekly Plague in London, 1640–1648



Some Plague Facts

- Plague epidemics recorded from Roman times to early 1900s.
- $\gtrsim 1/3$ Europe's population died in "Black Death" of 1348
 - ~ 300 years for the population to reach the same level.
- Recently (2011) established (at McMaster!) that the pathogen that caused The Black Death was *Yersinia pestis*

[Bos *et al.* 2011, *Nature* **478**, 506–510]

- More recently (2014) established (again at McMaster!) that the pathogen that caused The Plague of Justinian (541–543 AD) was *Yersinia pestis*

[Wagner *et al.* 2014, *Lancet Infectious Diseases* **14**, 319–326]

- *Y. pestis* still a concern?
Yes: Rodent reservoir, antibiotic-resistant strains, bioterrorism
- **Spatial data** for any plagues? Yes, for London in 1665...

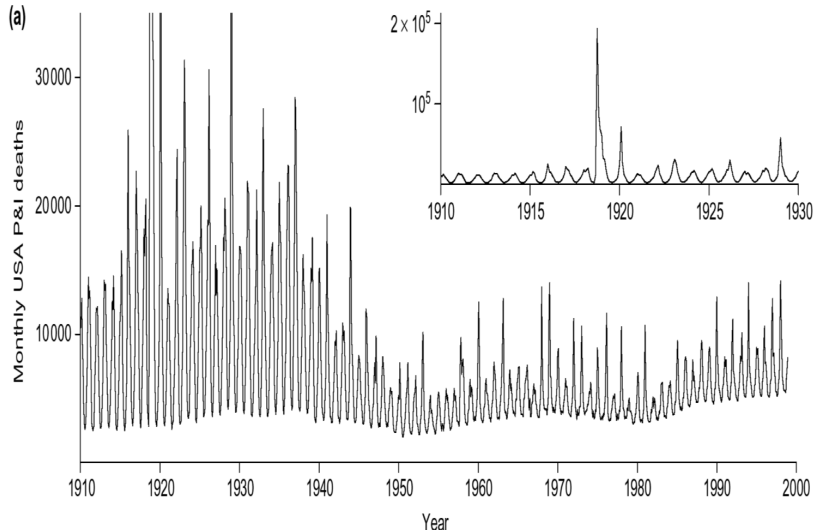
Visualization of spatial structure of Great Plague

- GIS encoding of parish boundaries
- Overlay parish boundaries on more modern map for reference
- Colour parishes as they become infected
- Is there evidence for spatial spread or was the spatial pattern random?
- DE low-tech animation...
- CBC high-tech animation...
 - *The Nature of Things*, 21 August 2014.
<https://www.imdb.com/title/tt8746582/>

Visualization of entire course of the Great Plague

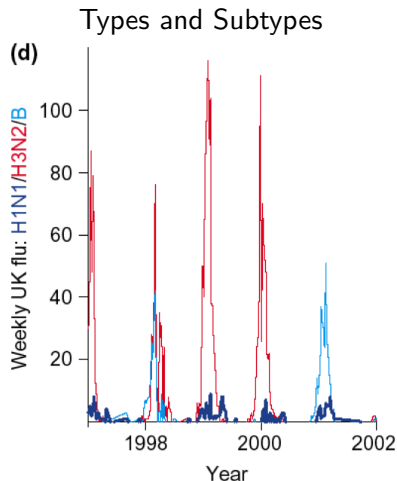
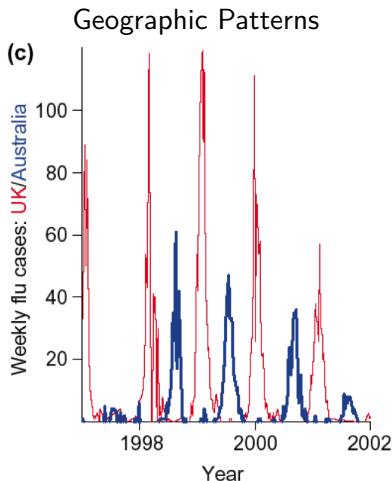
- What happened after initial spatial spread?
- Visualize full spatial epidemic structure
- Show magnitude of epidemic in each parish with cylinder.
- Epidemic Visualization (EpiVis) software by Junling Ma.
- Full spatial dynamics of the Great Plague (7 fps, 40 fps)
 - there is one frame per day, so 7 fps means *one week per second*

P&I mortality in U.S.A., 1910–1998



Earn, Dushoff & Levin 2002, *Trends in Ecology and Evolution* **17**, 334–340

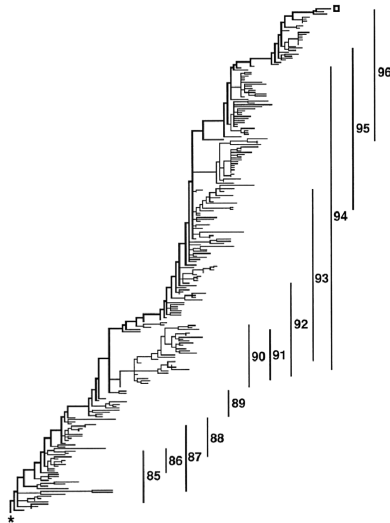
Influenza Incidence Patterns (lab confirmed)



Earn, Dushoff & Levin 2002, *Trends in Ecology and Evolution* **17**, 334–340

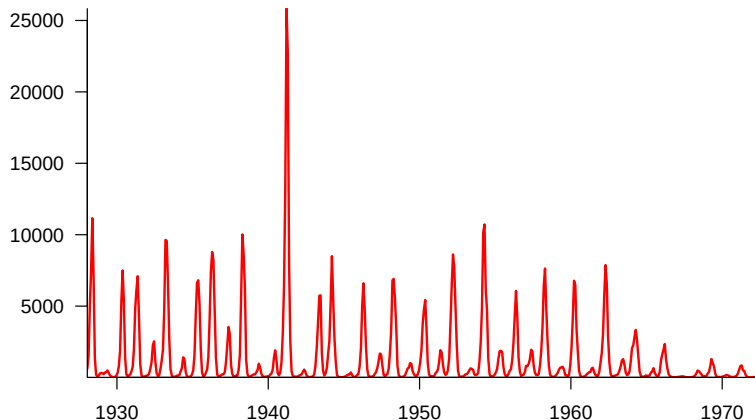
Influenza Evolution

Molecular
phylogenetic
reconstruction of
influenza A/H3N2
evolution,
1985–1996
(Fitch *et al.* 1997)



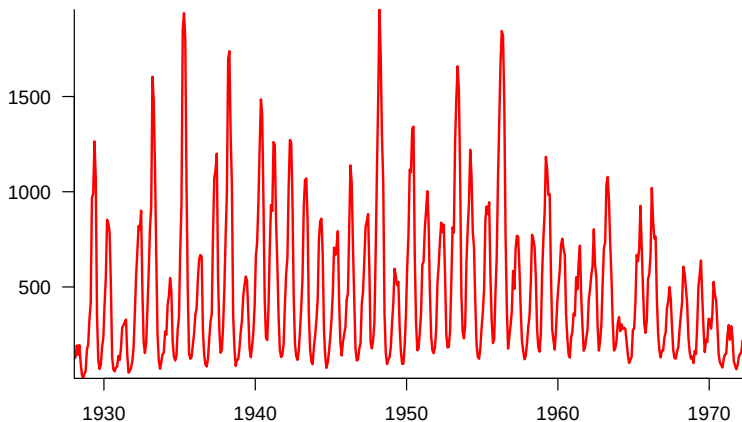
Measles in New York City, 1928–1972

Monthly Cases



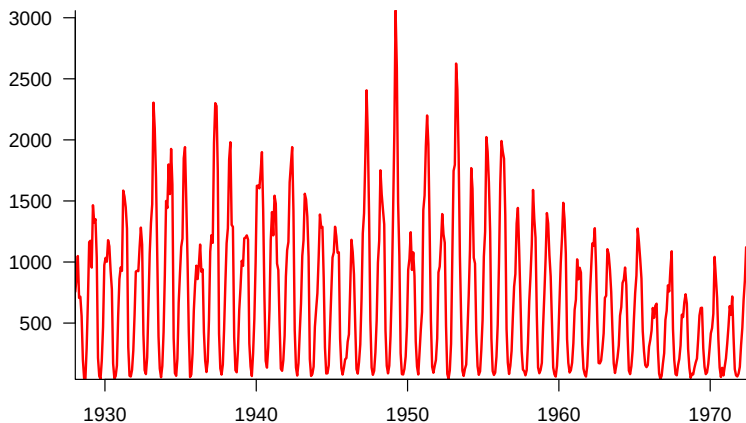
Mumps in New York City, 1928–1972

Monthly Cases

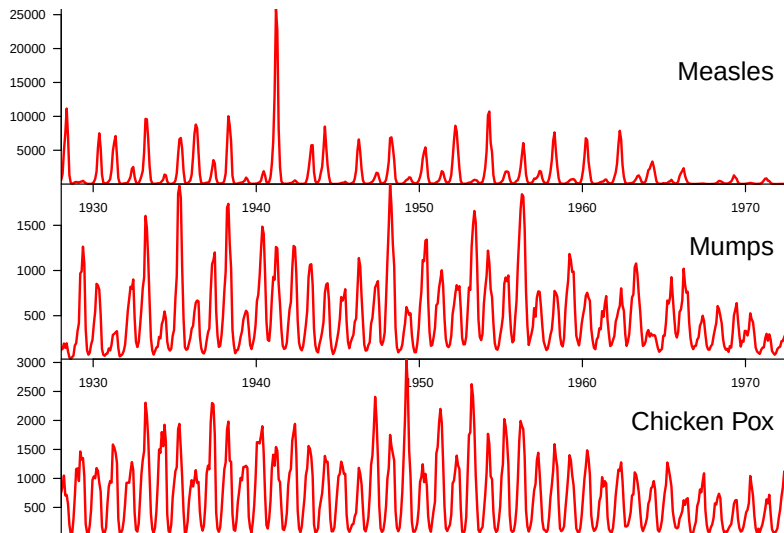


Chicken Pox in New York City, 1928–1972

Monthly Cases

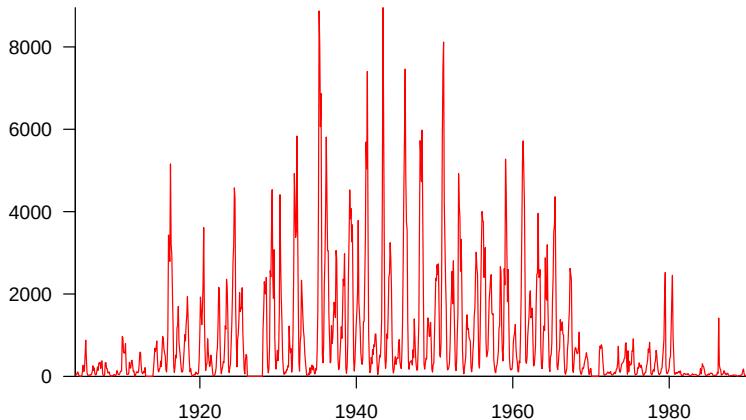


Childhood diseases in New York City, 1928–1972



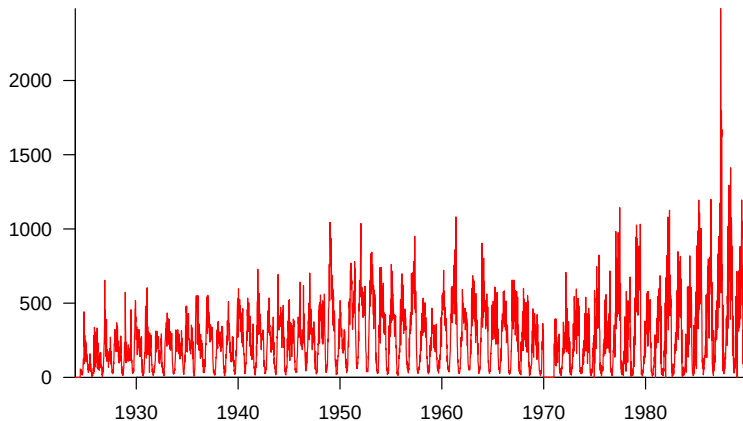
Measles in Ontario, 1904–1989

Monthly Cases



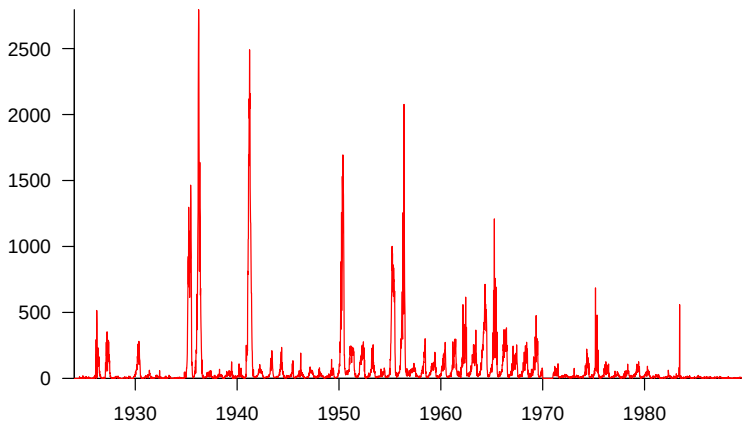
Chicken Pox in Ontario, 1924–1989

Monthly Cases



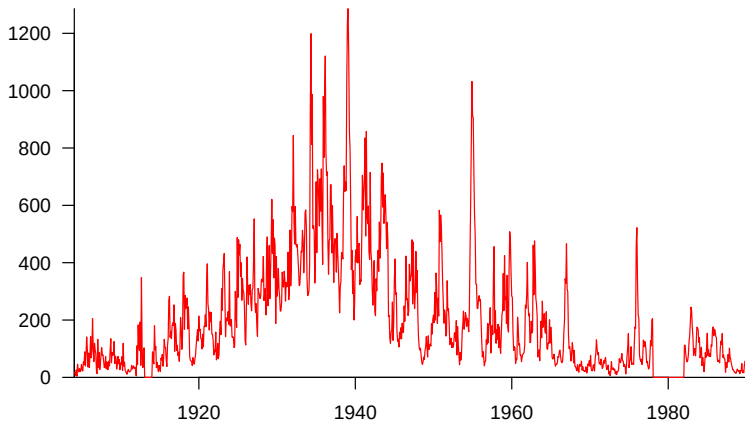
Rubella in Ontario, 1924–1989

Weekly Cases

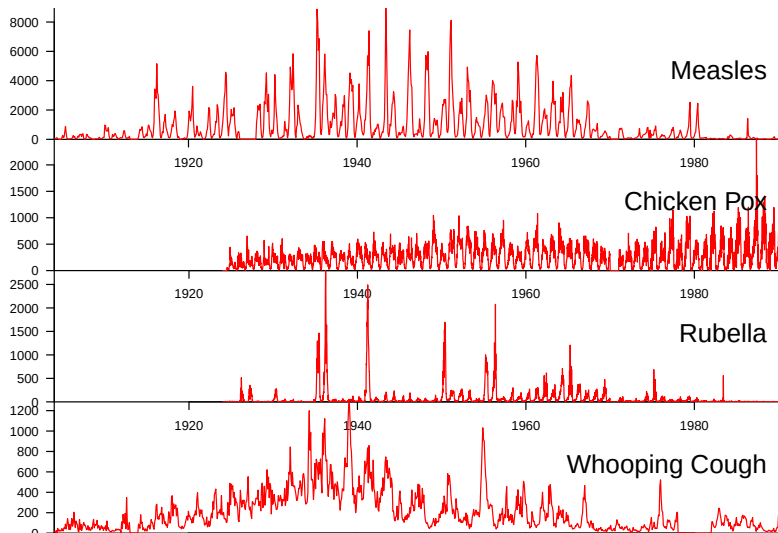


Whooping Cough in Ontario, 1904–1989

Monthly Cases



Childhood diseases in Ontario, 1904–1989



Ontario Disease Notification Data

Province of

YEAR: 1939

COUNTY

MUNICIPALITY

Month	Week End.	CSM		C.P.		DIP.		DYS. A/B		EN. LETH.		ERY.S.		G.C.		FLU.		INF. JAUN.		G.M.		MEAS.		MUMPS		PAR. TYPH.		
		C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	C	D	
Jan.	7	1		452	1	3	0	1	0				5	1	101	0	8	1	17	0	17	0	670	1	56	0	2	0
	14	2	2	1	490	0	8	0				5	0	82	0	21	1	18	0	18	0	850	0	92	0	1	0	
	21	3	2	1	511	0	9	3		0	1	5	0	89	0	16	2	26	0	22	0	932	0	98	0			
	28	4	1	0	384	0	2	0				2	0	73	0	164	0	10	0	28	0	933	1	24	0			
	Total		5	2	1837	1	22	3	1	0	0	1	17	1	352	0	203	4	71	0	85	0	3385	2	210	0	3	0
Feb.	4	5		355	0	7	1	1	0			3	0	83	0	57	1	24	0	25	0	1335	1	110	0	2	0	
	11	6	2	1	363	0	1	0	1	0		7	0	82	0	27	1	41	1	29	0	1033	0	91	0	1	0	
	18	7	2	1	354	1	2	0				4	1	68	0	103	1	35	0	44	0	1161	0	59	0			
	25	8	1	1	308	0	2	0				9	0	56	0	177	0	19	0	28	0	999	0	73	0			
	Total		5	3	1380	1	12	1	2	0		23	1	289	0	364	3	119	1	126	0	4588	1	333	0	3	0	
Mar.	4	9	1	1	271	0	7	1	3	1		7	0	93	0	1114	19	21	0	40	0	1131	2	169	0	1	0	
	11	10			239	0	7	0	2	0		8	1	61	0	1371	8	31	0	32	0	845	0	91	0	2	0	
	18	11			166	0						6	0	66	0	1322	6	5	0	59	0	969	2	69	0	1	0	
	25	12	1	2	236	0	1	0	1	0		7	0	63	0	806	16	9	0	20	0	879	0	170	0	2	0	
	Total		2	3	912	0	15	1	6	1		28	1	283	0	4623	49	66	0	151	0	3884	4	389	0	34	0	
Apr.	1	13	2	0	139	0	3	0	1	0		8	0	93	0	667	6	1	0	24	0	950	0	89	0	3	0	
	8	14	2	0	162	0	1	0	1	0		5	0	67	0	731	22			14	0	770	0	65	0	1	0	
	15	15	2	0	108	0	1	0			0	1	11	0	41	0	529	16	2	0	16	0	745	0	56	0		
	22	16	1	1	134	0	2	0	1	0	1	1	6	0	64	0	245	8	2	0	26	0	845	0	54	0		
	29	17	5	1	167	0	4	0	2	0	2	1	3	0	55	0	124	9	2	1	13	0	746	1	120	0		
Total		12	2	110	0	11	0	5	0	3	3	33	0	322	0	2314	61	9	1	99	0	4016	1	284	0	4	0	
	6	18	2	0	104	0	1	0	2	0		4	0	71	0	76	3	1	0	14	0	877	0	63	0	3	0	

Dominion Bureau of Statistics Disease Notification Data

VITAL STATISTICS BRANCH - COMMUNICABLE DISEASE SECTION

Cases of *H. influenzae* Reported by Provincial Health Departments, Year *1924*

WEEK ENDING	P.E.I.		N.S.		N.B.		QUE.		ONT.		MAN.		SASK.		ALTA.		B.C.		CANADA	
	WKS	MT	WKS	MT	WKS	MT	WKS	MT	WKS	MT	WKS	MT	WKS	MT	WKS	MT	WKS	MT	WKS	MT
<i>Jan</i> 5			11										1						12	
2 12			29										18						47	
3 19			37										32						69	
4 26			75	152			68		181		36	13	64		97		4	88	602	
<i>5 FEB</i> 2			12		1								53						66	
6 9			5										40						45	
7 16			31										14						45	
8 23			2	50	1	2	267		202		48	4	111		116		1	7	797	
<i>9 MAR</i> 1			2										21						23	
10 8													9						9	
11 15			3										11						14	
12 22			60										34						94	
13 29			2	67			144		140		52	15	90		15		7	17	515	
<i>14 APR</i> 5			9										11						20	
15 12			1										12						13	
16 19			26		1								8						35	
17 26			14	50	3	4	42		140		39	16	47		67		5	33	394	
<i>18 MAY</i> 3			26										2						28	

All Historical Canadian Infectious Disease Data

<https://canmod.net/digitization/>

A Century of Weekly Notifiable Disease Incidence Data by
Province in Canada

<https://doi.org/10.1371/journal.pgph.0005550>



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 4MB3/6MB3 Mathematical Biology

Instructor: David Earn

Lecture 4

Epidemic Data Tools II; Mechanistic Modelling of Recurrent
Epidemics I

Tuesday 29 Sep 2026

Announcements

■ Assignment 2:

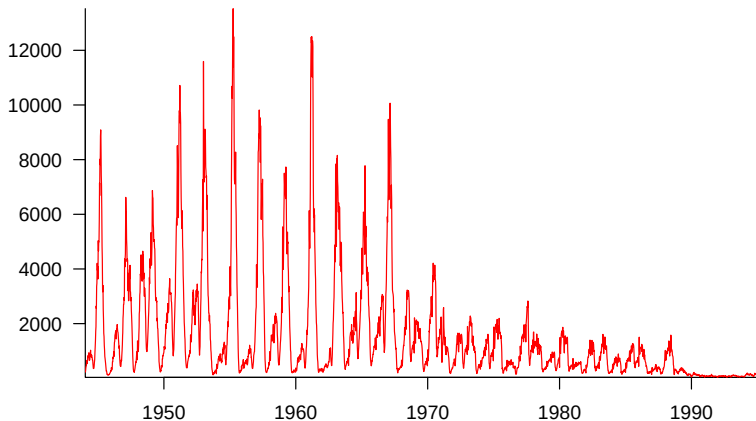
Due Thursday 8 October 2026 at 11:59 p.m. on crowdmark and (with source files) by e-mail.

Recurrent epidemics of childhood infections

- Childhood diseases in New York City, 1928–1972
- Childhood diseases in Ontario, 1904–1989

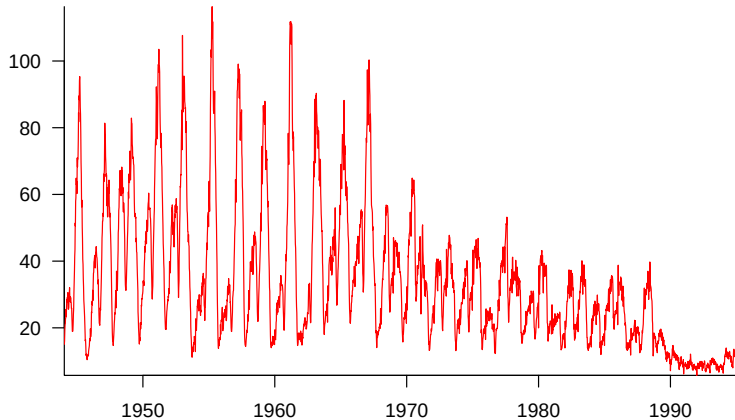
Measles incidence in England and Wales, 1944–1995

Weekly Cases



Measles incidence in England and Wales, 1944–1995

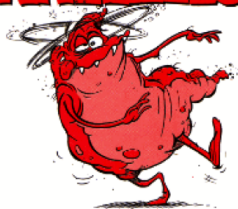
Sqrt(Weekly Cases)



Why study measles epidemics?

- $\sim 107,500$ deaths from measles in 2023
- A major cause of vaccine-preventable deaths.
- Potential impact in developed countries during vaccine scares (e.g., MMR scare in UK in 1990s).
- Understand past patterns
- Predict future patterns
- Manipulate future patterns
- Develop vaccination strategy that can...

**BRING
MEASLES
TO ITS
KNEEZLES!**



Other reasons to model infectious disease epidemics

- Mathematical models make hypotheses and inferences precise
 - Give better advice to policymakers
 - Make better predictions
- Host-pathogen dynamics are important aspects of ecosystem dynamics
 - Infectious disease models more likely to be successful than predator-prey models
- Excellent data for human infectious diseases
 - Models can be tested!

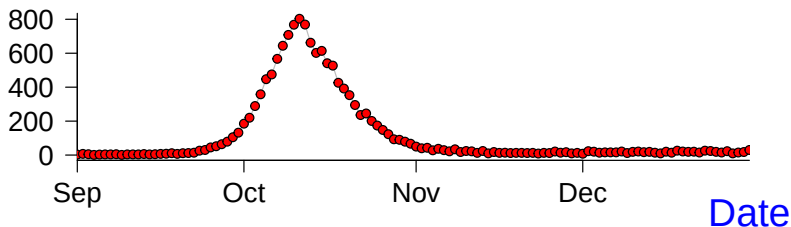
Modelling population dynamics of childhood infections

- The basic SIR model cannot explain recurrent epidemics.
- What should we do?...The usual options:
 - 1 Get depressed, drop the course.
 - 2 Keep developing models until we can explain recurrent epidemics.
- First, let's talk about tools that allow us to make our questions about time series data more precise.

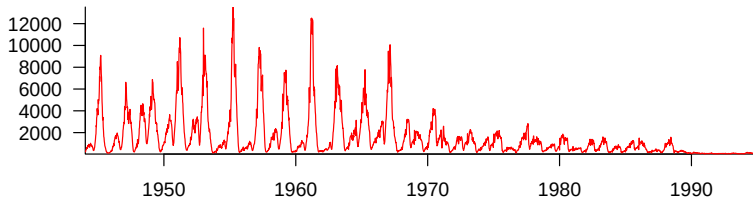
Epidemic Data Analysis

Time Plots of Temporal Epidemic Patterns

1918 P&I

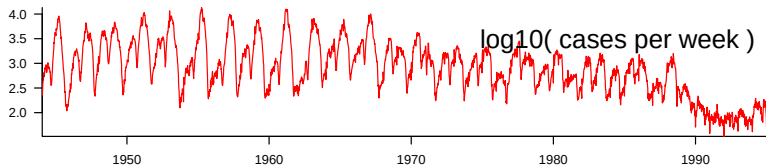
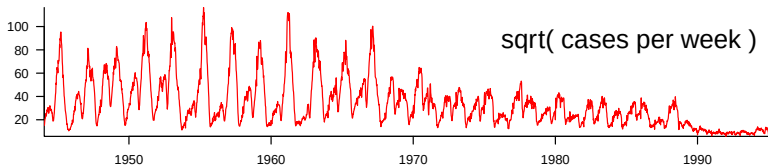
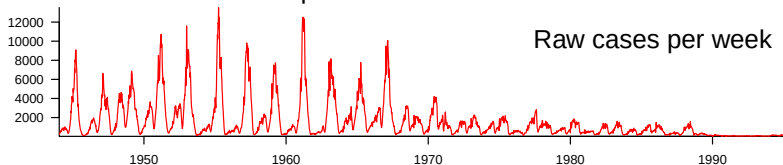


Weekly Measles in England and Wales



Time Plots of Transformed Data

- Reveal unobvious aspects of time series



Time Plots of Smoothed Data

- Reveal trends clouded by noise or seasonality

- *Moving Average:*

$$x_t \rightarrow \frac{1}{2a+1} \sum_{i=-a}^a x_{t+i}$$

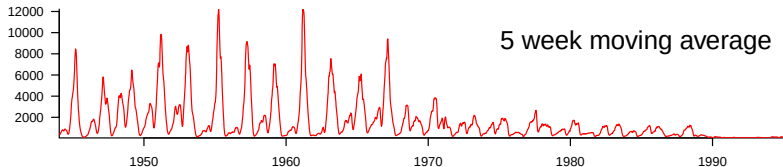
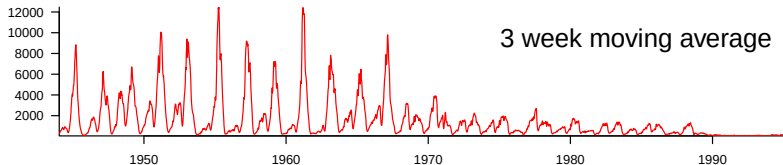
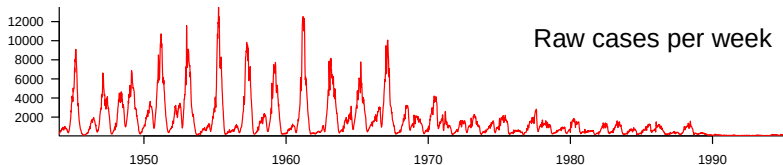
- Replace original data points x_t with averages of nearby points.

- *Linear filter:*

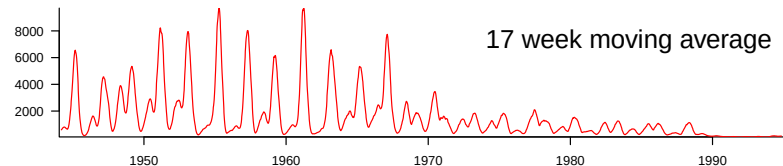
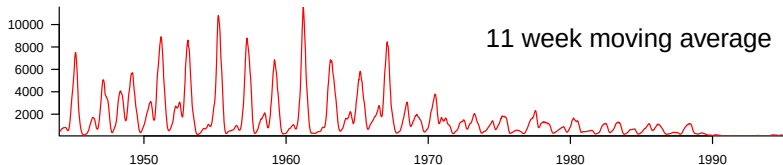
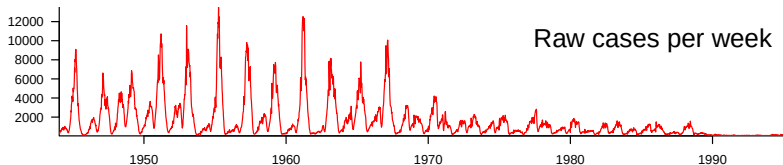
$$x_t \rightarrow \sum_{i=-\infty}^{\infty} \lambda_i x_{t+i}$$

- Generalization of moving average.
- *Weights* λ_i can be nonlinear functions of i .

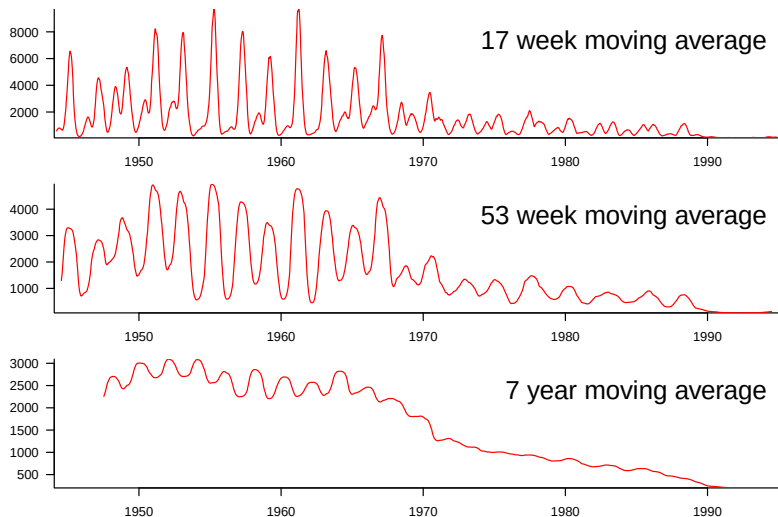
Time Plots of Smoothed Data



Time Plots of Smoothed Data



Time Plots of Smoothed Data



Correlation

- Recurrent epidemics $\implies$ number of cases now is correlated with number of cases in the past and the future.
- Given N pairs of observations of different quantities, $\{(x_i, y_i) : i = 1, \dots, N\}$, the *correlation coefficient* is defined to be

$$r = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2 \sum_{i=1}^N (y_i - \bar{y})^2}}$$

where $\bar{x}$ and $\bar{y}$ are the means of $\{x_i\}$ and $\{y_i\}$, respectively.

Correlation

Properties of the correlation coefficient:

- $-1 \leq r \leq 1$ (Proof? [Cauchy-Schwarz inequality](#))
- $r = 1 \iff$ all points lie on a line with positive slope (“complete positive correlation”)
- $r = -1 \iff$ all points lie on a line with negative slope (“complete negative correlation”)
- $r \simeq 0 \implies$ “uncorrelated”
- *Interpretation:* r^2 is the proportion of the variance in y explained by a linear function of x .

Derivations and discussions:

- [MathWorld on \$r^2\$](#) , [Wikipedia on \$r^2\$](#)
- [Wikipedia on general coefficient of determination](#)

Autocorrelation

- Given a single sequence of observations $\{x_t : t = 1, \dots, N\}$, we can compute the correlation of each observation with the observation k time steps in the future.
- Thus, we consider the pairs of observations $\{(x_t, x_{k+t}) : t = 1, \dots, N - k\}$ and define the *autocorrelation coefficient at lag k* to be

$$r_k = \frac{\sum_{t=1}^{N-k} (x_t - \bar{x}_{1,N-k})(x_{k+t} - \bar{x}_{k+1,N})}{\sqrt{\sum_{t=1}^{N-k} (x_t - \bar{x}_{1,N-k})^2 \sum_{t=1}^{N-k} (x_{k+t} - \bar{x}_{k+1,N})^2}}$$

where $\bar{x}_{1,N-k}$ and $\bar{x}_{k+1,N}$ are the means of first and last $N - k$ observations, respectively.

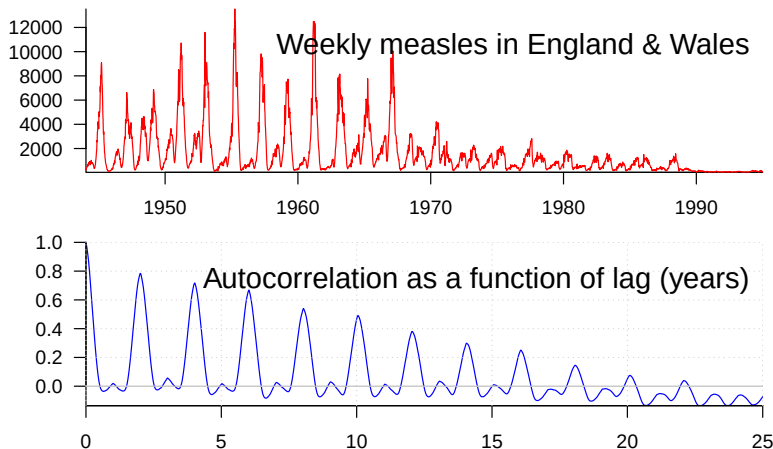
Autocorrelation

- If number of observations N is large and lag $k \ll N$ then

$$r_k \simeq \frac{\sum_{t=1}^{N-k} (x_t - \bar{x})(x_{k+t} - \bar{x})}{\sum_{t=1}^N (x_t - \bar{x})^2}$$

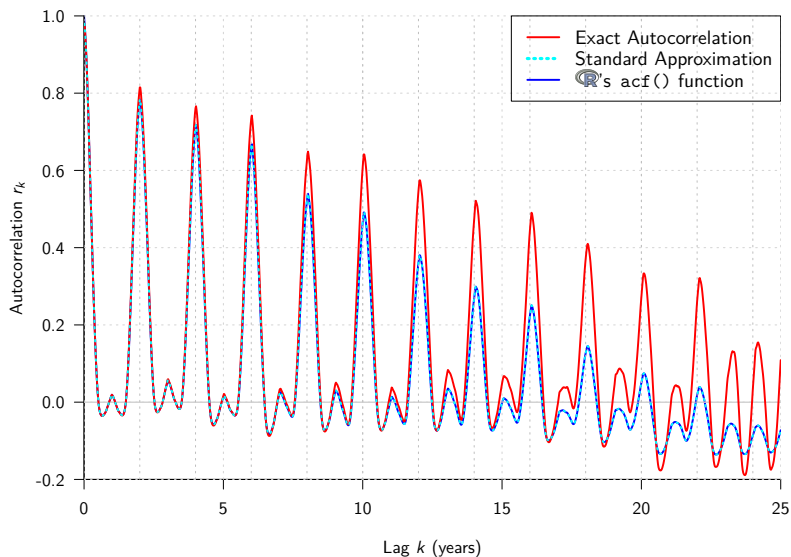
- Approximation of r_k is worse for larger lags k
- Plot of autocorrelation r_k as a function of lag k is called the *correlogram*.

Correlogram



- Peaks in correlogram $\implies$ periodicities in original time series.
- Correlograms of temporal segments are often informative.

Correlogram: exact vs. approximate r_k



Spectral Density

- Can we compute the dominant periods in the time series? (Rather than estimating them by eye from the [correlogram](#).)
- Express the time series as a [Fourier series](#):

$$x_t = a_0 + \left(\sum_{p=1}^{(N/2)-1} (a_p \cos \omega_p t + b_p \sin \omega_p t) \right) + a_{N/2} \cos \pi t,$$

where $\omega_p = 2\pi p/N$.

- Compute the [Fourier coefficients](#) $\{a_p\}$, $\{b_p\}$ by taking inner products with $\cos \omega_p t$ and $\sin \omega_p t$.

Spectral Density

- Fourier coefficients of x_t are:

$$a_0 = \bar{x} = \frac{1}{N} \sum_t x_t,$$

$$a_p = \frac{2}{N} \sum_t x_t \cos \omega_p t, \quad b_p = \frac{2}{N} \sum_t x_t \sin \omega_p t,$$

$$a_{N/2} = \frac{1}{N} \sum_t (-1)^t x_t,$$

where sum is over observation times.

- Estimated **power spectral density (PSD)** at frequency ω_p is*:

$$I(\omega_p) = \frac{N}{4\pi} (a_p^2 + b_p^2)$$

*The normalization by $N/4\pi$ is the convention chosen by [Chatfield \(2004, "Analysis of Time Series: An Introduction"\)](#). Other normalization conventions are also in common use.

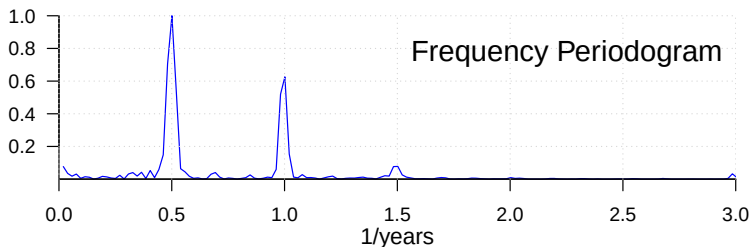
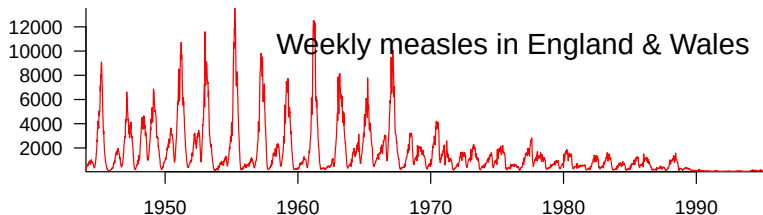
Spectral Density

- There are many different ways to express the **power spectral density** (aka **power spectrum**).
- Most common/useful equivalence is that the power spectrum is the **discrete Fourier transform** of the **correlogram**:

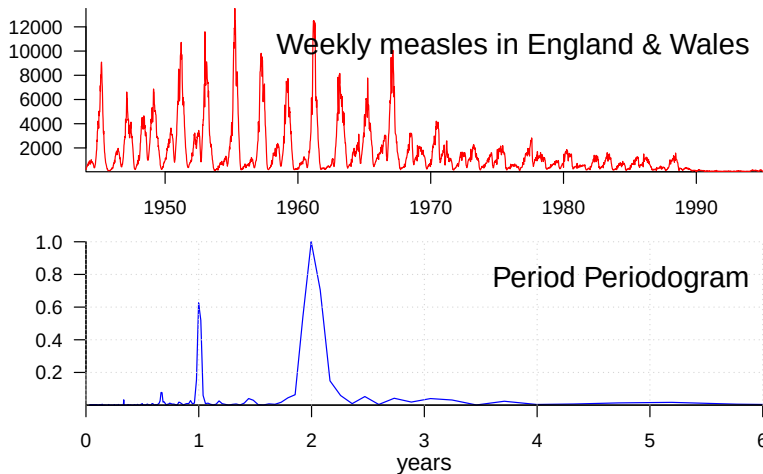
$$I(\omega_p) = \frac{1}{\pi} \left(r_0 + 2 \sum_{k=1}^{N-1} r_k \cos \omega_p k \right)$$

- Plot of estimated power spectrum as a function of frequency ω_p is called the **frequency periodogram** or just the **periodogram**.

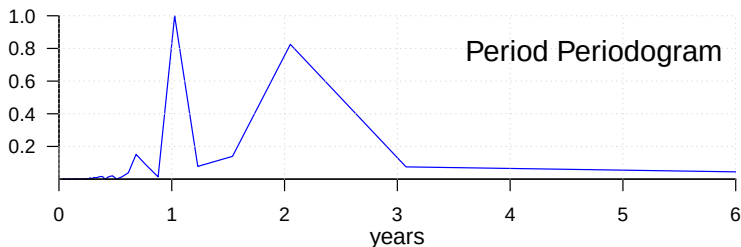
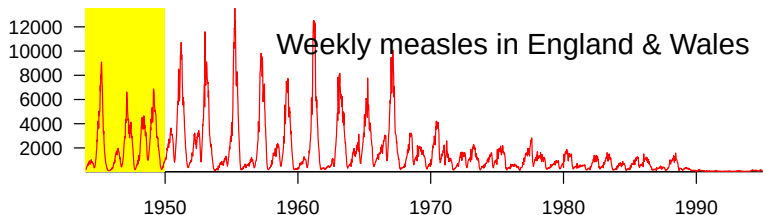
Spectral Density



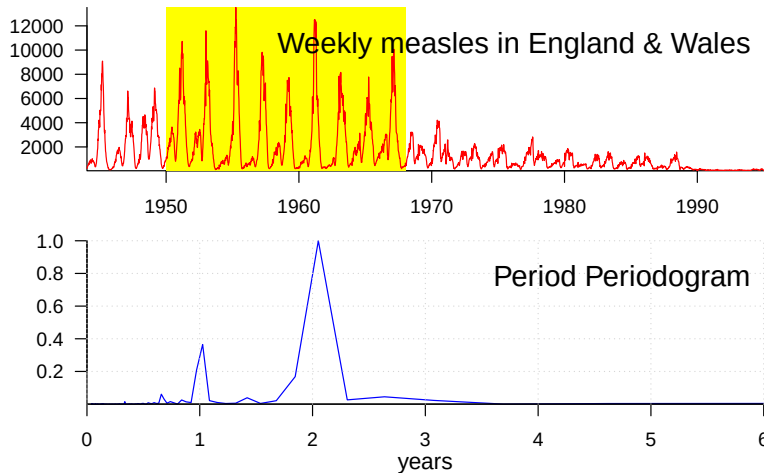
Spectral Density



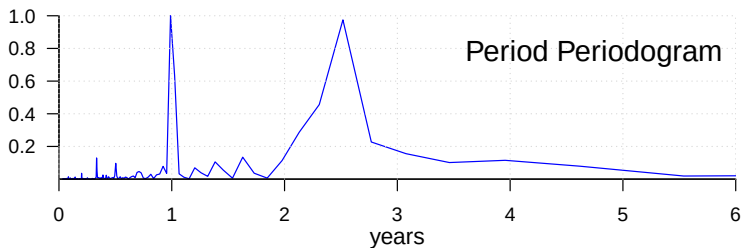
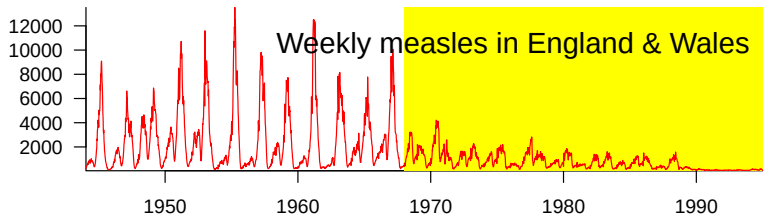
Spectral Density of Temporal Segments



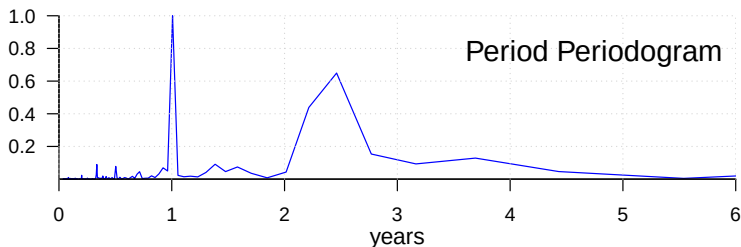
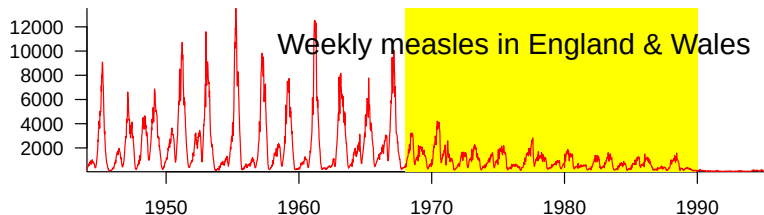
Spectral Density of Temporal Segments



Spectral Density of Temporal Segments



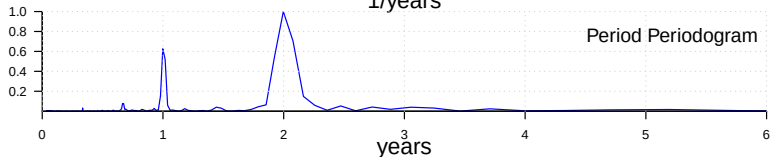
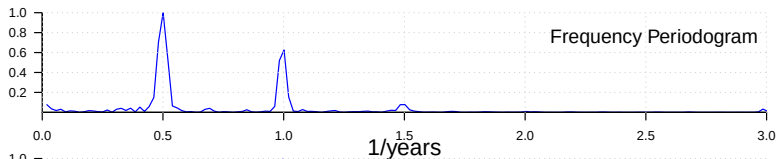
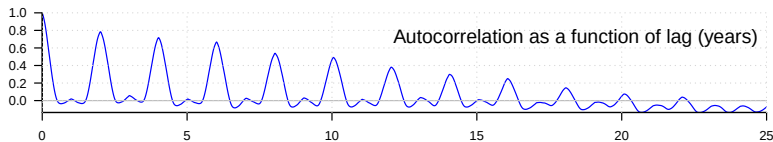
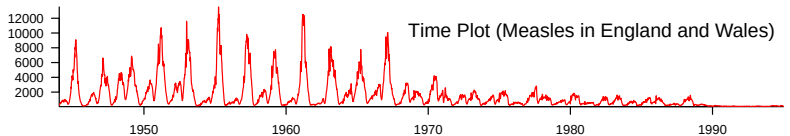
Spectral Density of Temporal Segments



Spectral Density Properties

- Periodogram is discrete Fourier transform of correlogram
- Same information in correlogram and periodogram
- Periodogram usually easier to interpret
- In $\mathbb{R}$, calculate power spectrum with `spectrum()`
- The power spectrum $I(\omega_p)$ partitions the variance in the time series with respect to frequency ω_p .
 - Parseval's theorem implies $\frac{1}{N} \sum_t (x_t - \bar{x})^2 = \frac{1}{2\pi N} \sum_{p>0} I(\omega_p)$.
But $\frac{1}{N} \sum_t (x_t - \bar{x})^2 = \text{Var}\{x_t\}$, hence $I(\omega_p)/(2\pi N)$ is the proportion of the variance in the time series associated with period $2\pi/\omega_p$.
[For details, see Chatfield (2004).]

Basic Time Series Analysis of Epidemic Data



More on Time Series Tools

Spectral Density of Temporal Segments

- Pre-war measles
- Post-war pre-vaccination measles
- Vaccination era measles
- Vaccination era measles until 1990

Time series analysis functions

 has built-in tools for time series analysis:

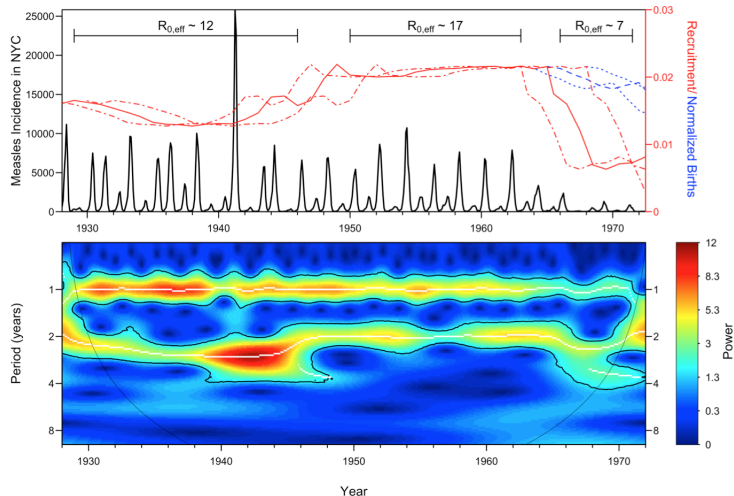
- Time plot: `plot()` *etc.*
- Linear filter (e.g., moving average): `filter()`
- Correlogram (auto-correlation function): `acf()`
- Periodogram (power spectrum): `spectrum()`

You will use all of these functions in **Assignment 4**.

More sophisticated spectral method

- Traditional power spectrum measures frequency content of entire time series.
- **Wavelet decomposition** is local in time.
 - Reveals changes in the spectrum over time without having to identify distinct temporal segments yourself.
 - Nice intro to wavelet analysis of time series:
Torrence and Compo (1998) "A Practical Guide to Wavelet Analysis" *Bulletin of the American Meteorological Society* **79**, 61–78
 -  packages for wavelet analysis of time series (e.g., [WaveletComp](#), [wavelets](#)), and at least one [book on wavelet methods in !\[\]\(2872da22d27cce77d1c954ccf8830414_img.jpg\)](#)

Wavelet Spectrum of Monthly Measles in New York City



Krylova & Earn 2013, *J. R. Soc. Interface* **10**, 20130098

Wavelet Spectrum of Weekly Measles in New York City

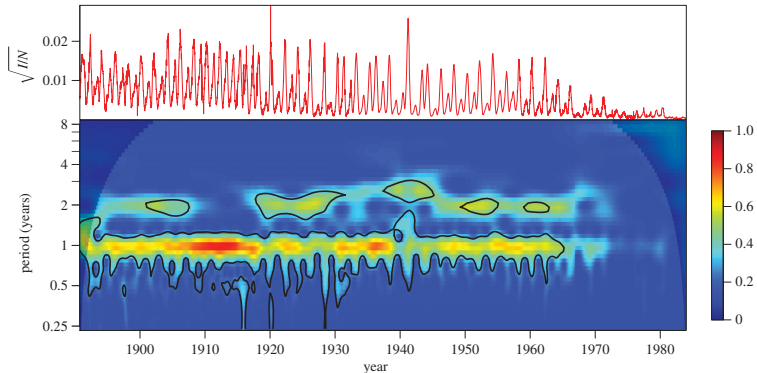
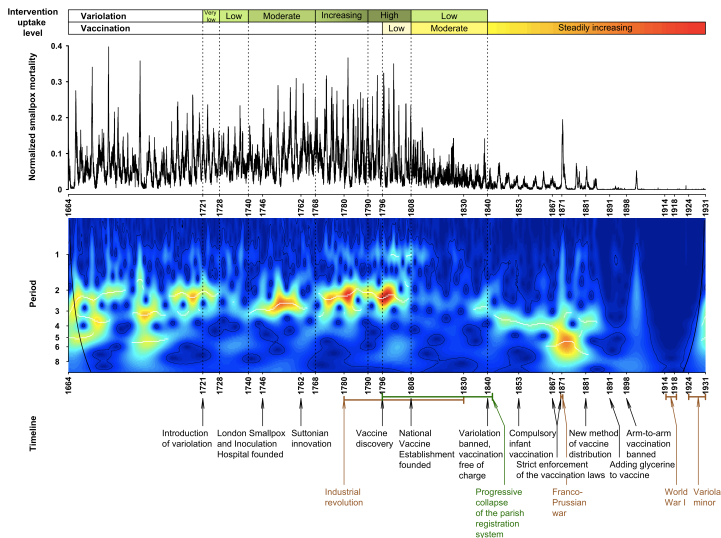


Figure 5. Observed measles dynamics in NYC from 1891 to 1984. (a) Square root of measles case reports, normalized by total concurrent population. (b) Colour depth plot of a continuous wavelet transform of the square root of normalized observed NYC measles cases (colour warmth scales with spectral power and 95% significance contours are shown in black). Shaded regions in the upper left and right indicate the cone of influence.

Hempel & Earn 2015, *J. R. Soc. Interface* **12**, 20150024

Wavelet Spectrum of Weekly Smallpox in London



Krylova & Earn 2020, *PLoS Biology* 18(12):e3000506

Statistical Modelling of Time Series

Statistical Modelling of Time Series

- Imagine time series $\{X_t\}$ is generated by random processes.
- Simplest case: X_t (number of cases at time t) is simply a random variable with a known distribution,

$$X_t = \mu + Z_t \quad (*)$$

where μ = time average number of cases
and $\{Z_t\}$ = sequence of random variables with zero mean.

- Might be a reasonable model for importation of new, infectious individuals into a focal community.
- Bad model for epidemics: ignores transmission from one individual to another.
 - There must be a correlation between the number of individuals in the focal community who are infected now and the number who will be infected in the near future.

Statistical Modelling of Time Series: AR and MA

- So, imagine that that successive data points in $\{X_t\}$ are correlated.
- For example, perhaps the data are generated by an *autoregressive (AR) process*:

$$X_t - \mu = \alpha_1(X_{t-1} - \mu) + \alpha_2(X_{t-2} - \mu) + \cdots + \alpha_p(X_{t-p} - \mu) + Z_t,$$

where the α_i are constants that determine the degree of correlation along the time series.

- Alternatively, the data might be generated by a *moving average (MA) process*:

$$X_t - \mu = \beta_0 Z_t + \beta_1 Z_{t-1} + \cdots + \beta_q Z_{t-q},$$

where the β_i are constants that define a weighted average.

Statistical Modelling of Time Series: ARMA

- More generally, the data might be generated by an *autoregressive moving average* “ARMA(p, q)” process:

$$X_t - \mu = \alpha_1(X_{t-1} - \mu) + \alpha_2(X_{t-2} - \mu) + \cdots + \alpha_p(X_{t-p} - \mu) \\ + \beta_0 Z_t + \beta_1 Z_{t-1} + \cdots + \beta_q Z_{t-q}.$$

Statistical Modelling of Time Series: ARIMA

- Finally, an *autoregressive integrated moving average* “ARIMA(p, d, q)” model allows for some non-stationarity by including weighted differences:

$$\begin{aligned} X_t - \mu = & \alpha_1(X_{t-1} - \mu) + \alpha_2(X_{t-2} - \mu) + \cdots + \alpha_p(X_{t-p} - \mu) \\ & + \gamma_1(X_{t-1} - X_{t-2}) + \gamma_2(X_{t-2} - X_{t-3}) + \cdots \\ & + \beta_0 Z_t + \beta_1 Z_{t-1} + \cdots + \beta_q Z_{t-q}. \end{aligned}$$

- The “I” in ARIMA refers to the fact that the stationary model involving differences must be “integrated” to obtain a (possibly non-stationary) model of the original data.

“Such a model is called an ‘integrated’ model because the stationary model which is fitted to the differenced data has to be summed or ‘integrated’ to provide a model for the non-stationary data.” (Chatfield, §3.4.6, p.42)

What kind of process generated our data?

- *How can we tell if our data were generated by such a process?
Can we identify an $AR(p)$, $MA(q)$ or $ARMA(p, q)$ process?*
- Compare time plots of these processes with time plot of our data? (Comparison by eye often challenging/unreliable.)
- Compare autocorrelation functions (correlograms) of these processes with correlogram of our data? (Better.)
- Compare power spectra (periodograms) of these processes with periodogram of our data? (Even better.)
- Compare wavelet spectra of these processes with wavelet spectrum of our data? (Better yet.)

Statistical Modelling of Time Series: ARMA fitting

- Looking at the power spectra of ARMA models would be instructive.
- But is there a better approach to discovering if an ARMA model could explain our data?
- Find the *best fit* ARMA parameters by minimizing the residual sum of squares. e.g., for an AR model, minimize:

$$S = \sum_{t=p+1}^N [(x_t - \mu) - \alpha_1(x_{t-1} - \mu) - \cdots - \alpha_p(x_{t-p} - \mu)]^2.$$

- More generally, we can find the best fit parameters of an ARIMA(p, d, q) model
 - Non-trivial, but there are standard methods
- Compare models with [Akaike Information Criterion \(AIC\)](#), which penalizes models that have more parameters
 - See [Earn \(2009\)](#) review article for more discussion of this.

Time series tools discussed so far...

- Statistical description of time series:
time plot, moving average, correlation coefficient,
autocorrelation, correlogram, power spectral density (PSD),
periodogram, wavelet spectrum
- Time series models:
AR, MA, ARMA, ARIMA

Statistical Modelling of Time Series

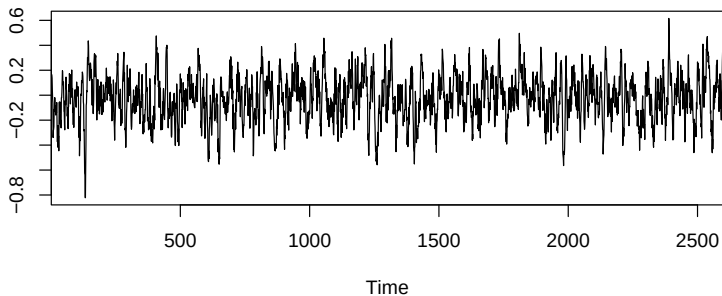
How to do it in  ...

- Simulate any $ARIMA(p, d, q)$ model with `arima.sim()`
- Fit an AR model to a time series with `ar()`
- Fit an ARIMA model to a time series with `arima()`
- Alternatively, there are specialized time series modelling packages.

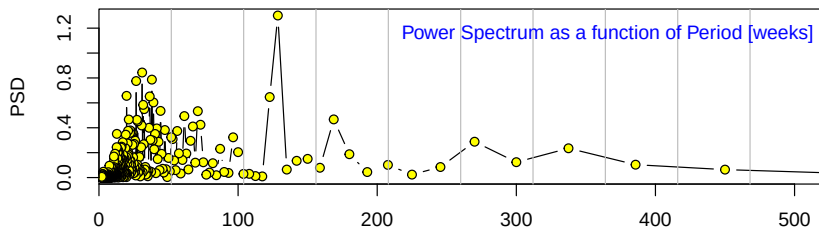
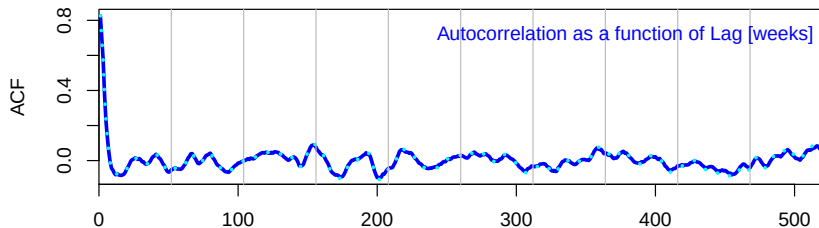
ARMA Example (50 years of weekly data)

```
my.model <- list(ar=c(1,-0.5,0.5,-0.25),ma=c(-0.25,0.5))  
my.sim <- arima.sim(n=52*50,model=my.model,sd=0.1)  
plot(my.sim,main="ARMA Example",ylab="",xaxs="i")
```

ARMA Example



ARMA Example (ACF and PSD up to 10 year lag)



Statistical Modelling of Time Series: Forecasting

- Once we have a fitted model, we can then use it to *forecast* future observations
- *Validate* this procedure by using part of the data to fit the model and then forecast the remainder of the data (*cf.* [cross-validation](#))
- How successful is this likely to be for an infectious disease time series?
 - Conceivably good for [chicken pox in NYC](#).
 - Less likely to be good for [measles](#)...at least for the main patterns...
 - One of the project options is to look at this more carefully.

Statistical Modelling of Time Series: Limitations

- It might be best to remove mean, trend and seasonality before fitting an ARMA model
 - But this means we will remove the aspects of the data about which we care most!
- The fitted parameters of an ARMA model have no obvious biological meaning
 - The model completely ignores any understanding we have of infectious disease transmission
- Statistical models use the time series itself to parameterize an ARMA (or more general) process
 - It would be better to have a model that we can parameterize from independently collected data and then see if that model can explain the observed time series

Mechanistic Mathematical Modelling

- SIR and all that...
- Takes into account transmission process...
- So why did we just spend time talking about statistical modelling of time series?
 - Important to be familiar with time series models that are in common use.
 - Helps us appreciate the value of mechanistic modelling.
 - Some processes that affect disease dynamics might be better modelled as ARMA or similar processes.
 - Weather (e.g., perhaps model $\beta = \beta(t)$ as an ARMA process)
 - Immigration
 - Ruling out an ARMA model (or at least one with a modest number of parameters) is a step towards finding a good model.
 - A combination of mechanistic and time series models could be useful.